

COMP9334

Capacity Planning for Computer Systems and Networks

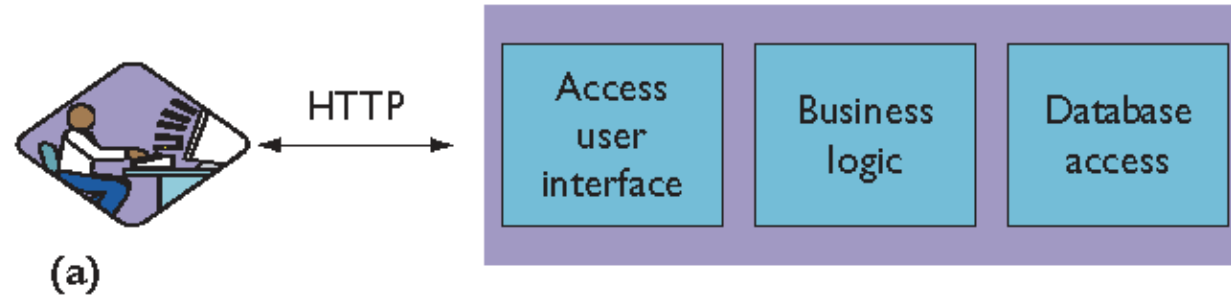
Week 8A: Fork-join queues

This lecture

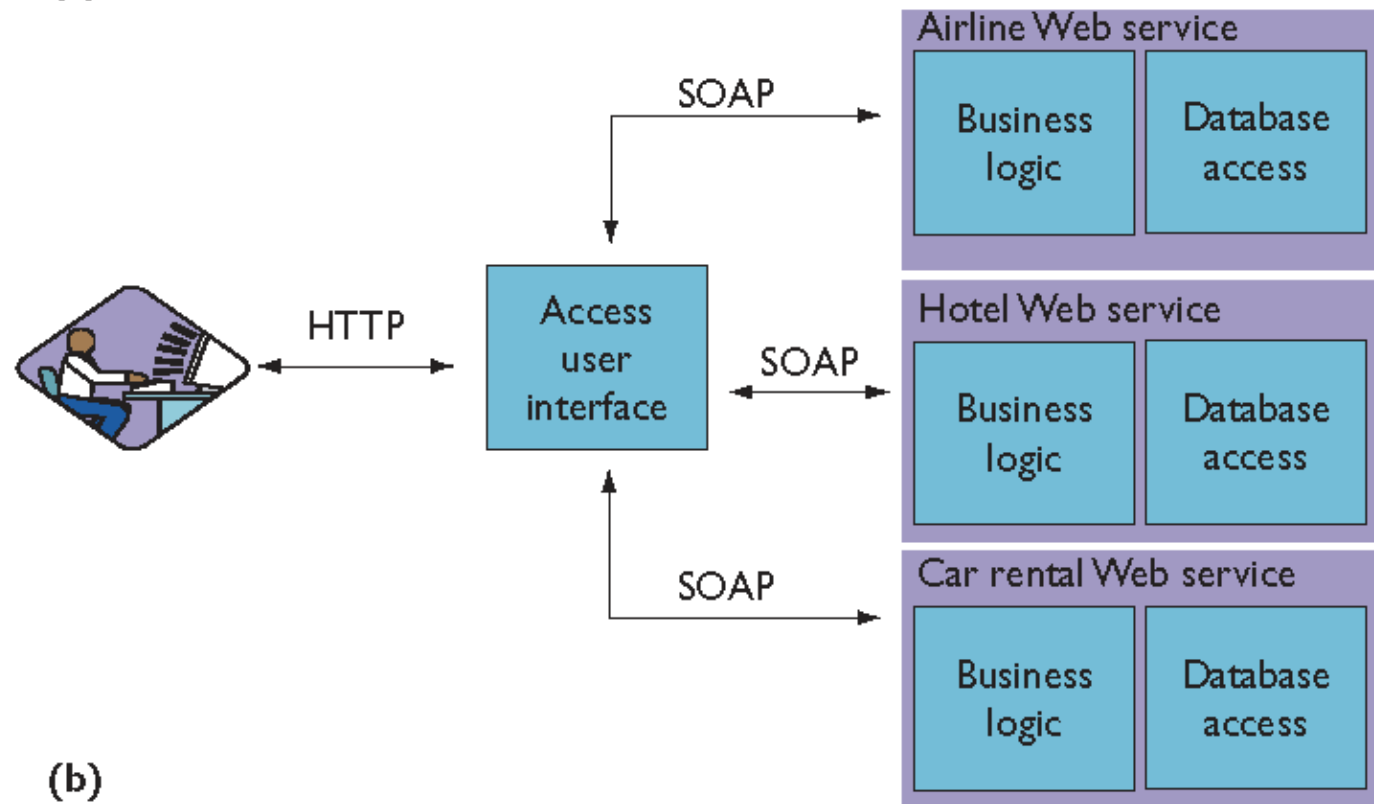
- Fork-join queue
 - Markov chain
 - MVA
- Fork-join queues can be used to model some parallel processing tasks in computer systems
 - An example is web services
 - Will discuss more examples later on
- Web services
 - What is it?
 - Performance analysis

Web access versus Web services

(a) Web access



(b) Composite web service for travel



Web service performance issues

- Metrics
 - Response time
 - Throughput
 - Availability
- Performance analysis method
 - Operational analysis
 - Markov chain

Web service flow graph

V_a, V_h, V_c :
Relative
Visit
Ratio

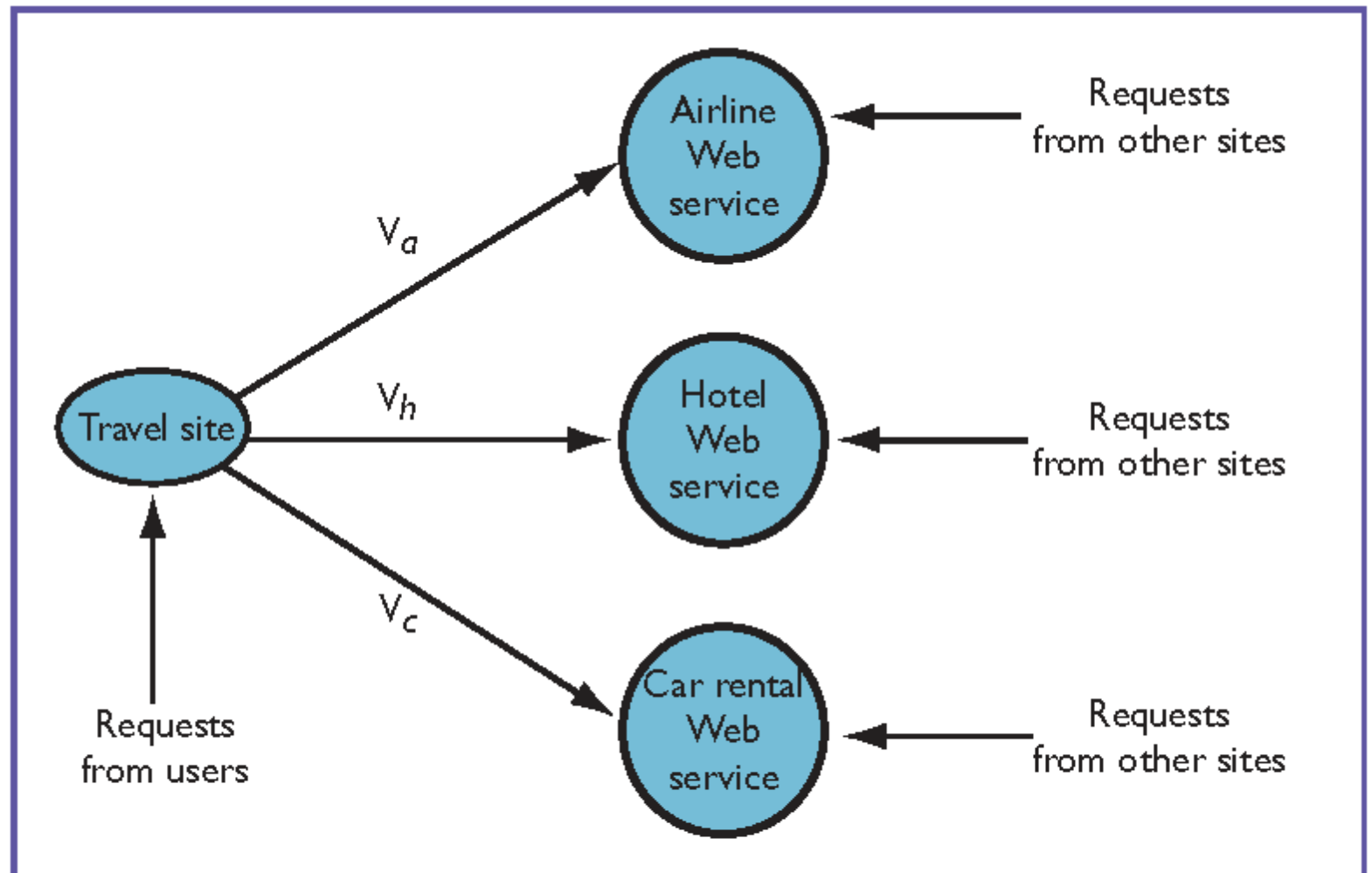
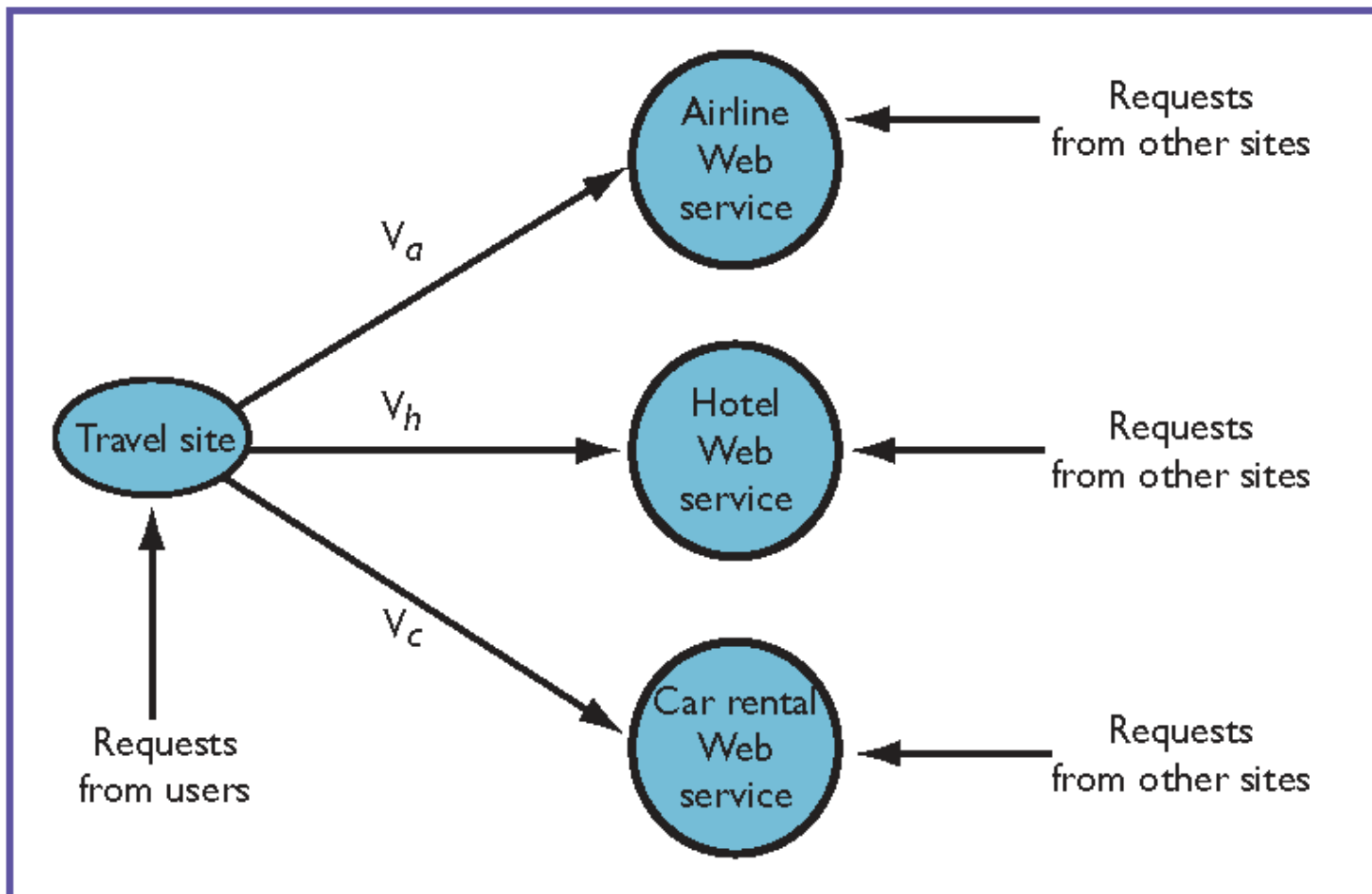


Figure 2. Web service flow graph. Arrows link the travel site to other Web services. The labels on the links indicate the average number of times a Web service is invoked per request to the travel site.

Every request to the travel site generates on average V_a requests to the Airline web service etc.



X_{TA} = Throughput of travel site

X_a = Throughput of airline Web service

$$X_a \geq V_a \times X_{TA}$$

Similarly,

$$X_a \geq V_a \times X_{TA}$$

$$X_h \geq V_h \times X_{TA}$$

$$X_c \geq V_c \times X_{TA}$$

X_h = Throughput of hotel
web service

X_c = Throughput of car
rental web service

- Can you find an upper bound on the throughput of the travel site

$$X_{TA} \leq \min\left\{\frac{X_a}{V_a}, \frac{X_h}{V_h}, \frac{X_c}{V_c}\right\}$$

Example:

$X_a = 20$ requests/s

$X_h = 15$ requests/s

$X_c = 10$ requests/s

$V_a = 4, V_h = 2, V_c = 1$

$$X_{TA} \leq \min\left\{\frac{20}{4}, \frac{15}{2}, \frac{10}{1}\right\} = 5 \text{ requests/s}$$

The airline web service is the bottleneck of the travel web site.

More complex web service graphs

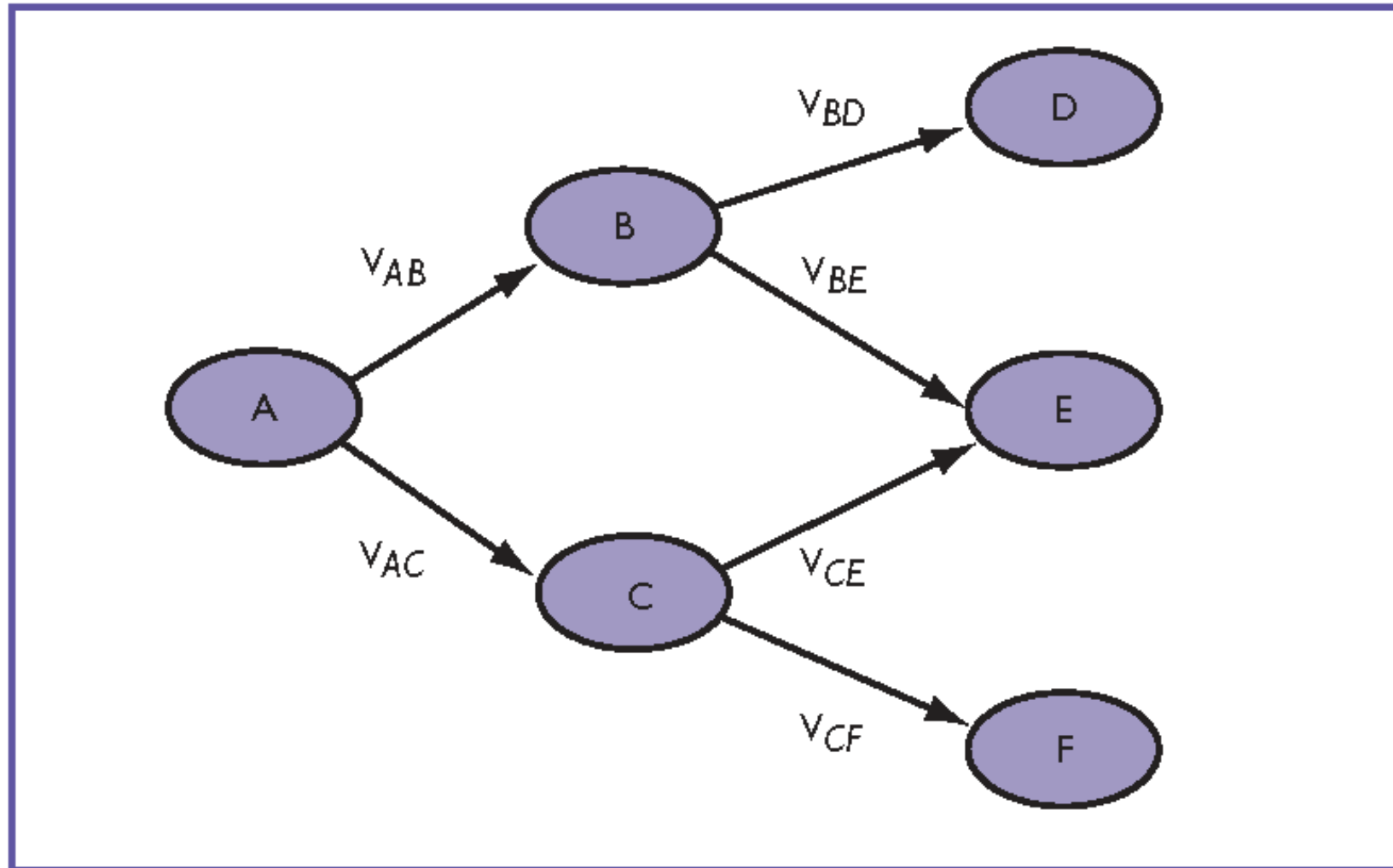


Figure 3. A more complex Web services flow graph. Web service A uses Web services B and C; B uses D and E; and C uses E and F.

What is the bound on throughput of web service A?

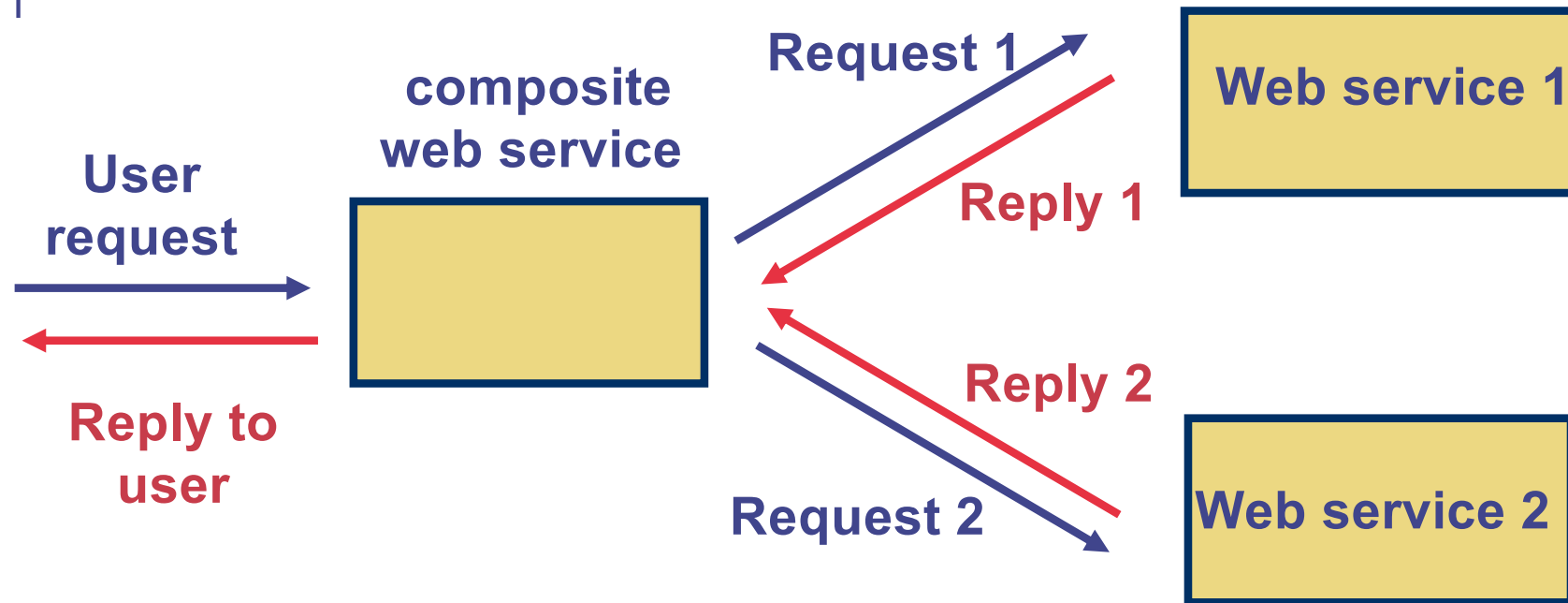
Bound on the throughput of web service A is:

$$X_A \leq \min \left\{ \frac{X_B}{V_{AB}}, \frac{X_C}{V_{AC}}, \frac{X_D}{V_{AB}V_{BD}}, \frac{X_F}{V_{AC}V_{CF}}, \frac{X_E}{V_{AB}V_{BE} + V_{AC}V_{CE}} \right\}$$

Response time analysis

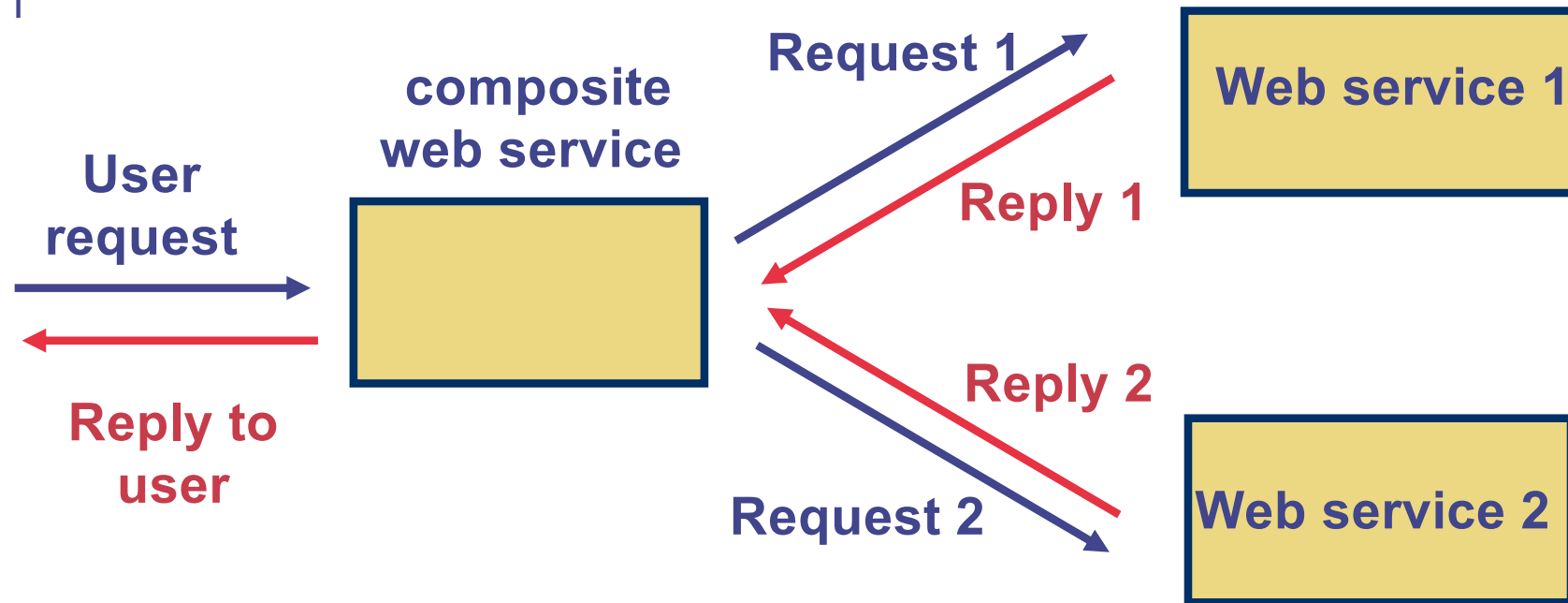
- The bottleneck analysis only gives an upper bound on the throughput
- Can we find the response time?
 - Markov chain
 - Approximate MVA
- We begin with a motivating example

A simple web service scenario (1)



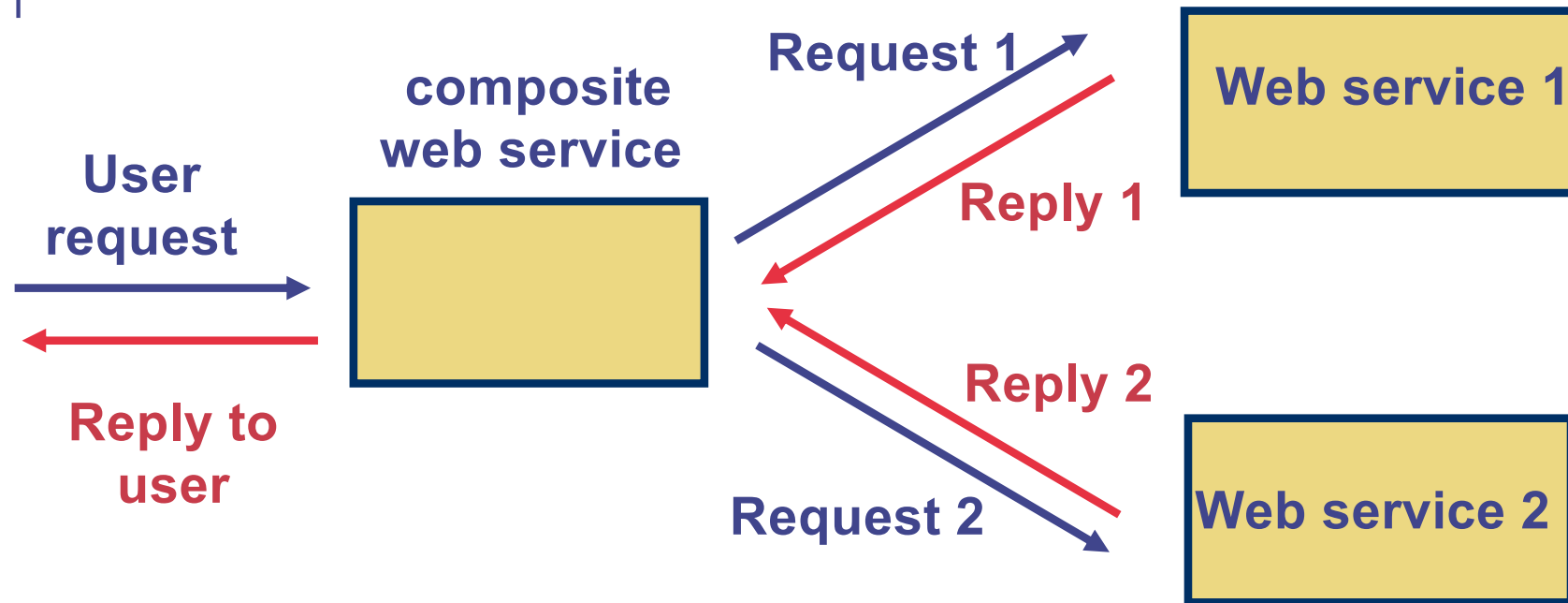
- A composite web service uses two web services
- Sequence of events
 1. Composite web service receives a user request
 2. Composite web service sends Request 1 and Request 2
 3. The web services reply *independently*
 - *That is, Reply 1 and Reply 2 may arrive at different times*
 4. After the composite web service receives *both* replies, it responds to the user

A simple web service scenario (2)



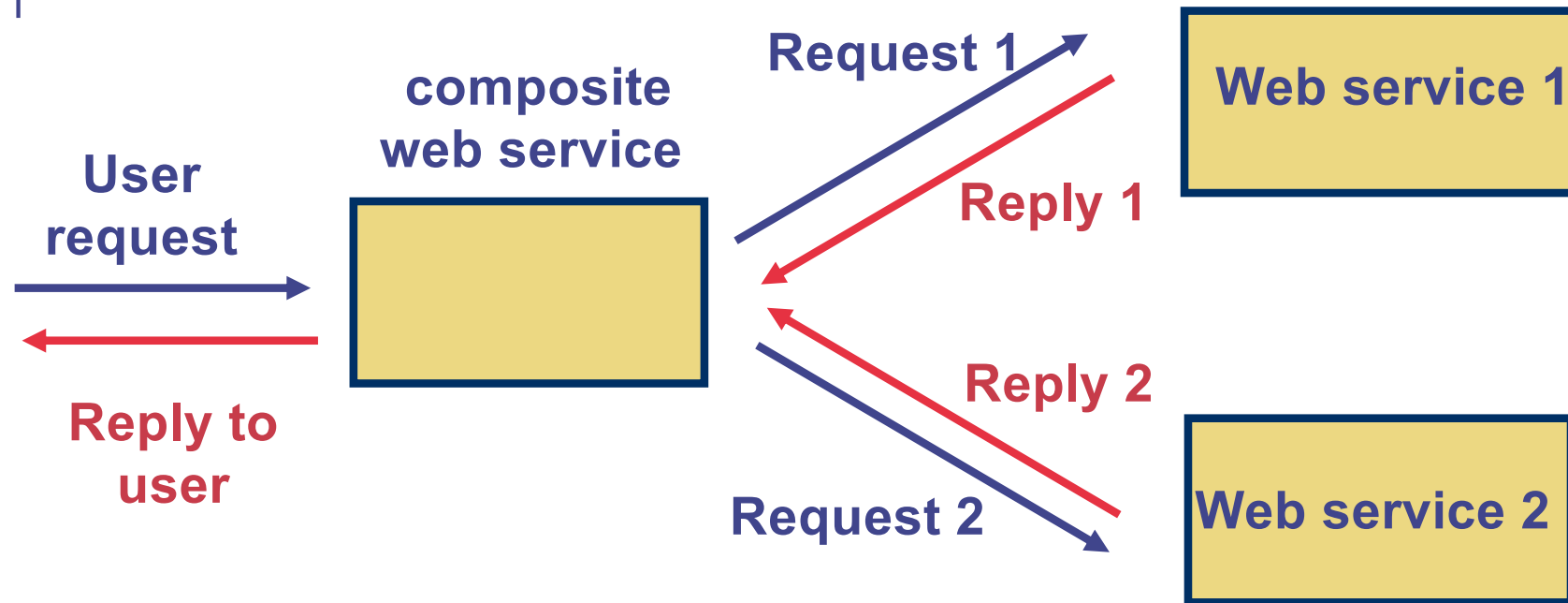
- Recall the definition of response time
- Response time of Web Service 1
= Time at which composite web service receives Reply 1 *minus*
Time at which composite web service sends Request 1
- Similarly for Web Service 2.

A simple web service scenario (3)



- Assuming that:
 - Web service 1 has a response time distribution of
 - 0.2s with probability 0.5
 - 0.3s with probability 0.5
 - Web service 2 has a response time distribution of
 - 0.2s with probability 0.5
 - 0.3s with probability 0.5
- What is the average time that the composite web service has to wait until both replies are returned?

A simple web service scenario (4)



- What if the service time distribution is:
 - Web service 1 has a response time distribution of
 - 0.2s with probability 0.5
 - 0.3s with probability 0.5
 - Web service 2 has a response time distribution of
 - 0.2s with probability 0.5
 - 0.5s with probability 0.5
- What is the average time that the composite web service has to wait until both replies are returned?

Analysis scenario

- Lesson learnt: Slow web services can become the bottleneck for composite web service
- We consider Composite Web Services (illustration next slide)
 - With parallel invocation of N services
 - Web services 1 through $N-1$ have a mean service time of S (exponentially distributed)
 - Web service N has a mean service time of $\frac{1}{\alpha} \times S$ (exponentially distributed)
 - The next service step can only be completed after all these N steps have been completed.

Note that if $\alpha < 1$, then server N is slower than the other $(N-1)$ servers.

Servers 1 to N-1 : mean response time = S

Server N: mean response time = $\frac{1}{\alpha} \times S$

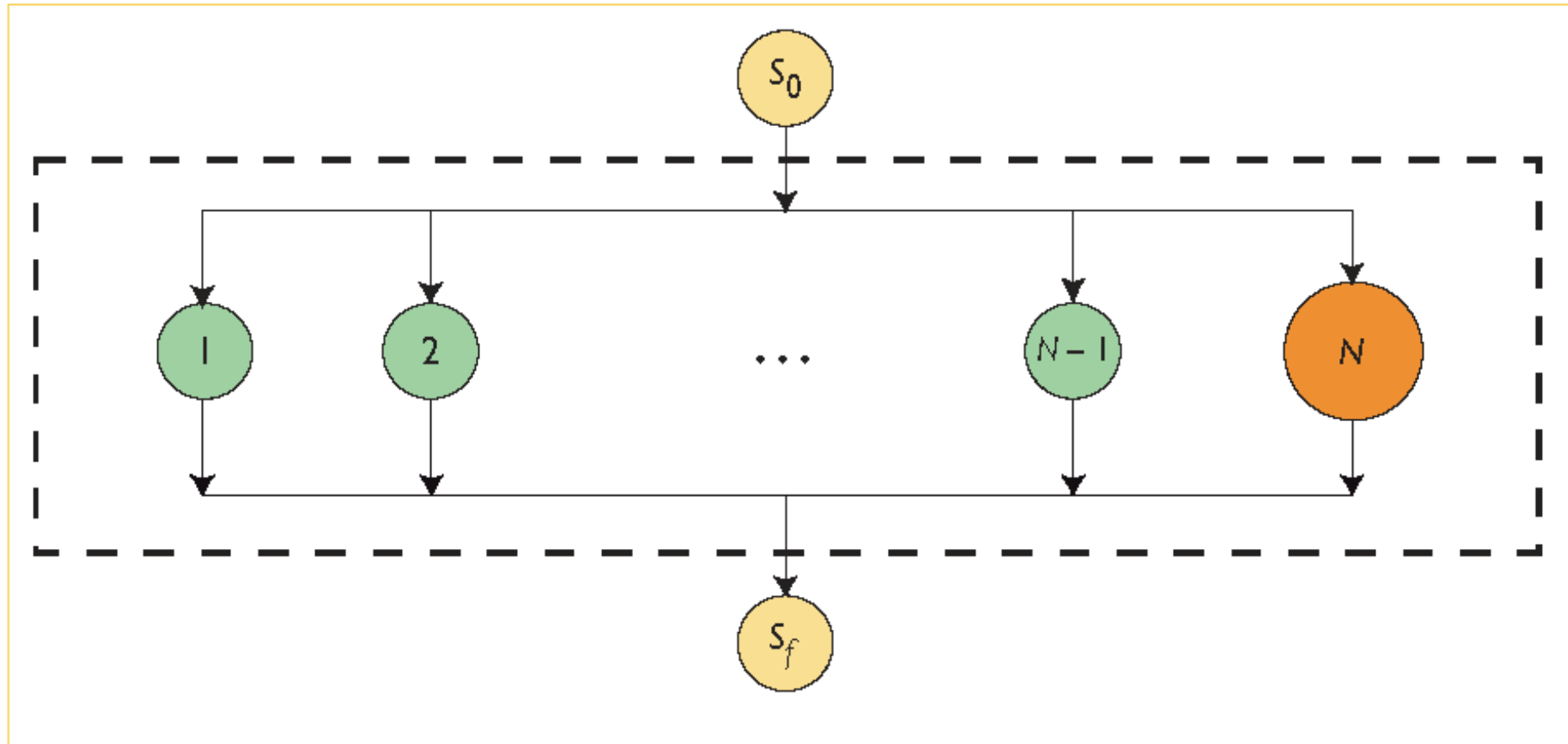


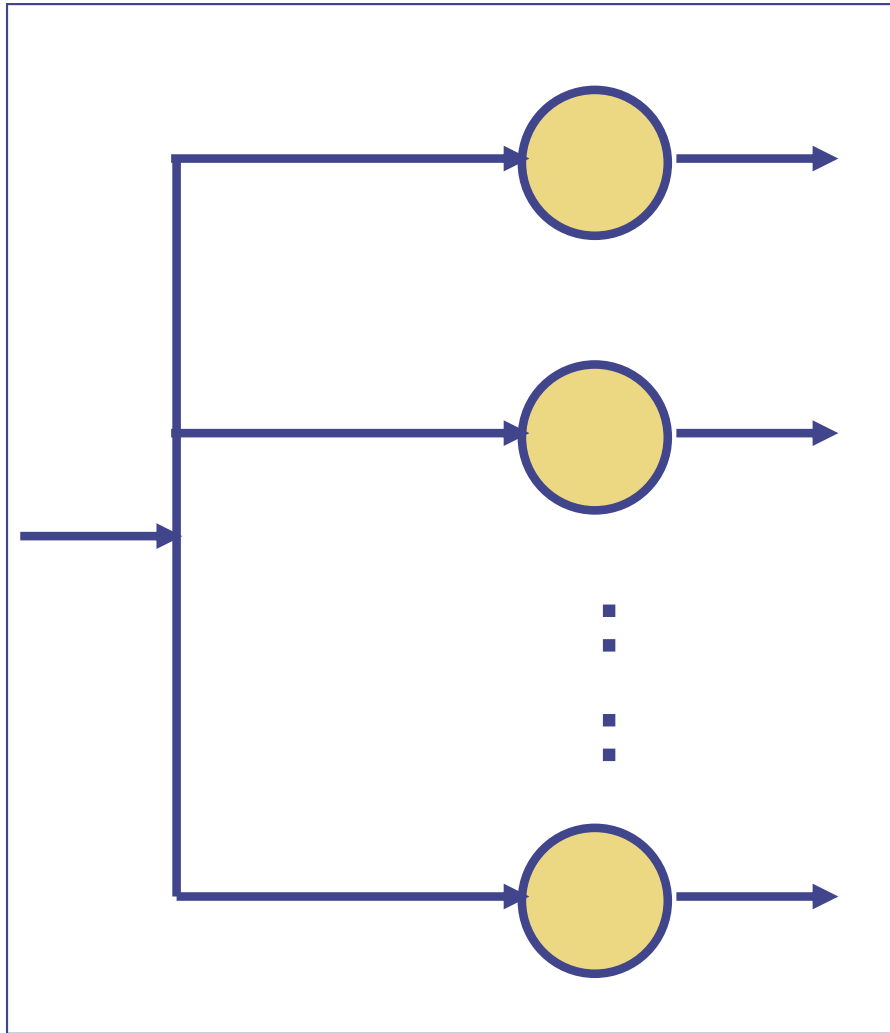
Figure 1. A composite Web service. After an initialization step S_0 , N Web services are invoked in parallel. Service N takes longer than the others, and the final step S_f can only be carried out after all N services have completed.

Fork-join system

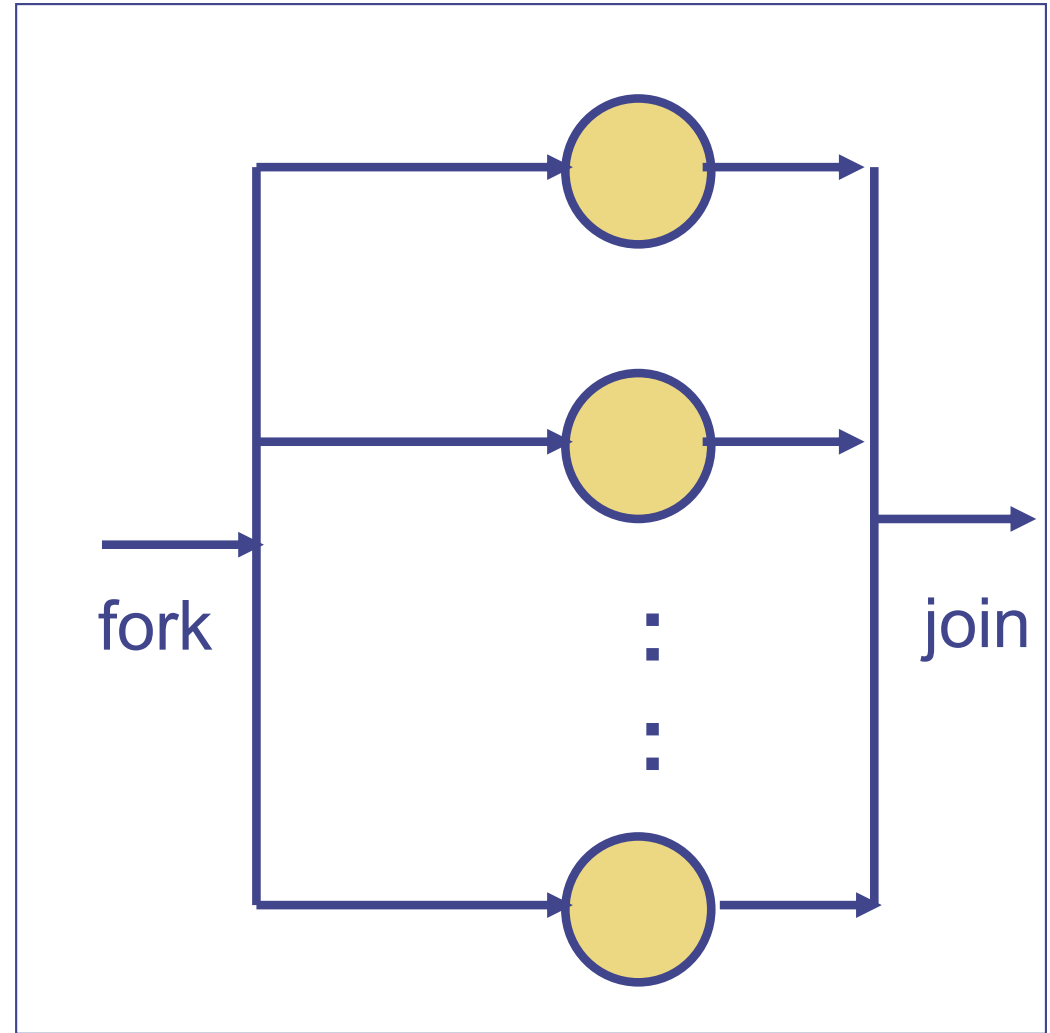
- The type of system described earlier is known as fork-join system
 - Fork is referring to the parallel invocation
 - All services must complete at the joining point before the next service can start

You've seen multi-server systems before:

M/M/m queue



Fork-join queue



What is the difference between these two queueing networks?

M/M/m type of multi-server versus fork-join

- M/M/m

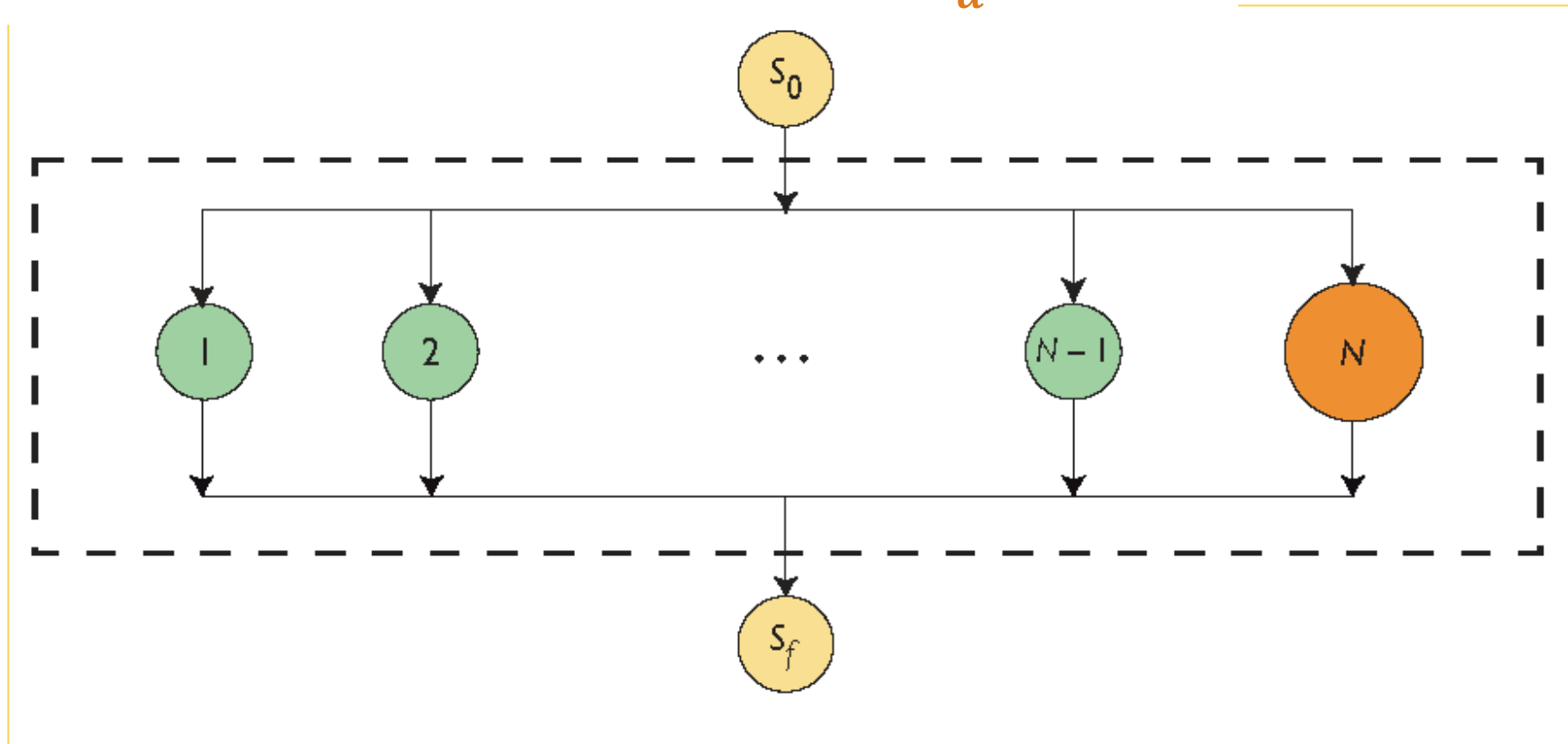
- An arriving request will be served by only one server
- Once a request has been completed by the server, the request leaves permanently

- Fork-join

- An arriving request will generate multiple sub-tasks where each sub-task will be processed by a server
- The results of the sub-tasks have to be joined together
- Need to wait for the results of all the sub-tasks before joining can proceed

Servers 1 to N-1 : mean response time = S

Server N: mean response time = $\frac{1}{\alpha} \times S$



- We want to understand how α affects the response time of the composite web services
- Let $T(\alpha)$ = response time as a function of α

What is $T(1)$?

- In this case, all constituent web services have the same response time distribution
- If all mean response times are exponentially distributed with mean S

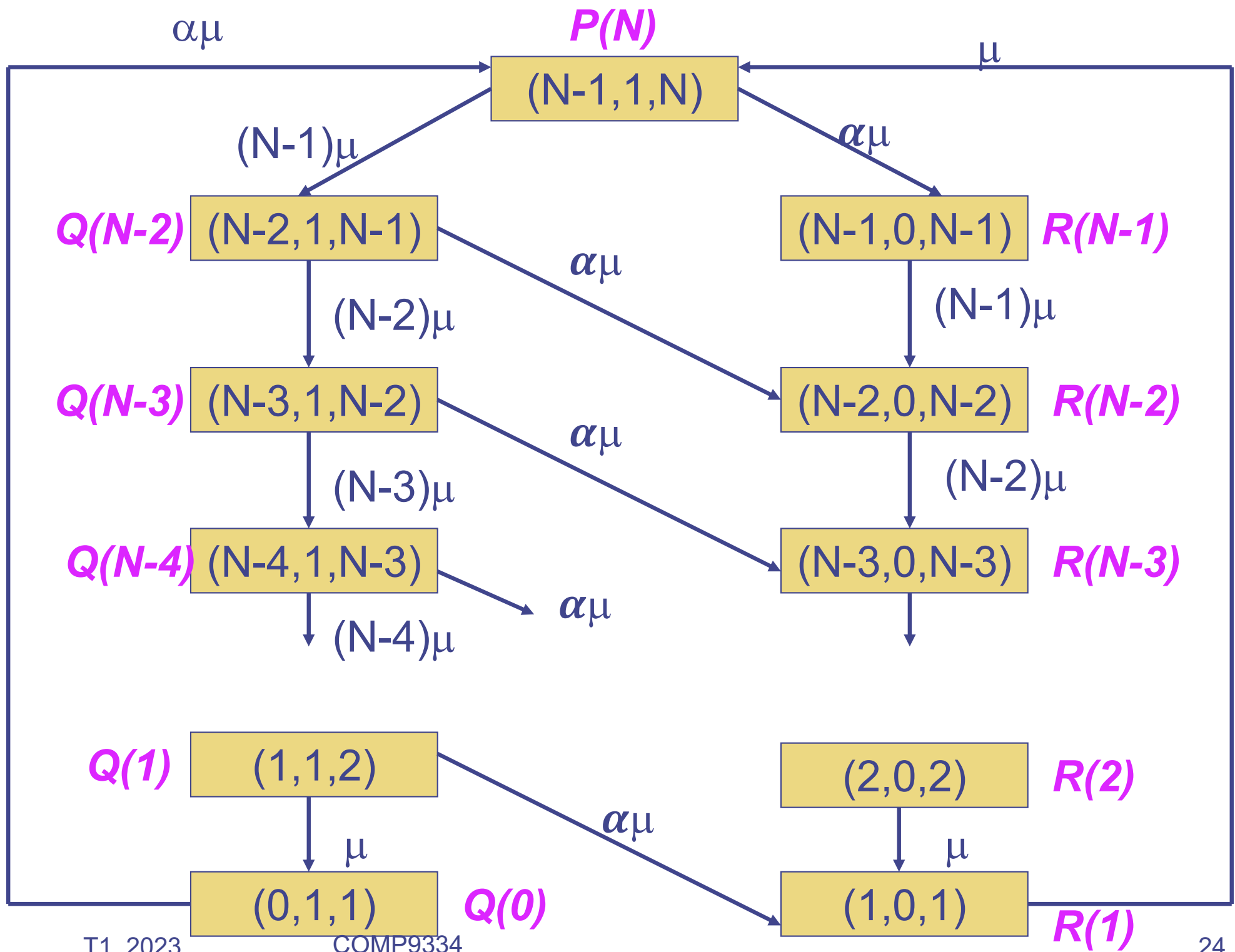
$$T(1) = \underbrace{\left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{N}\right)}_{=H_N} S$$

$H_N = N$ -th harmonic number

(We will explain how this is obtained later.)

How about $T(\alpha)$ for $\alpha < 1$?

- We use Markov chain.
- States (i,j,k)
 - i ($i = 0, \dots, N-1$) is the number of web services still running in fast Web services
 - j ($j = 0, 1$) is the number of web services running on the slow Web service
 - k ($k = 1, 2, \dots, N$) is the number of web services yet to complete
- Define $\mu = \frac{1}{S}$



$$T(\alpha) = \frac{S}{\left(N - 1 + \alpha\right) P(N)}$$

where

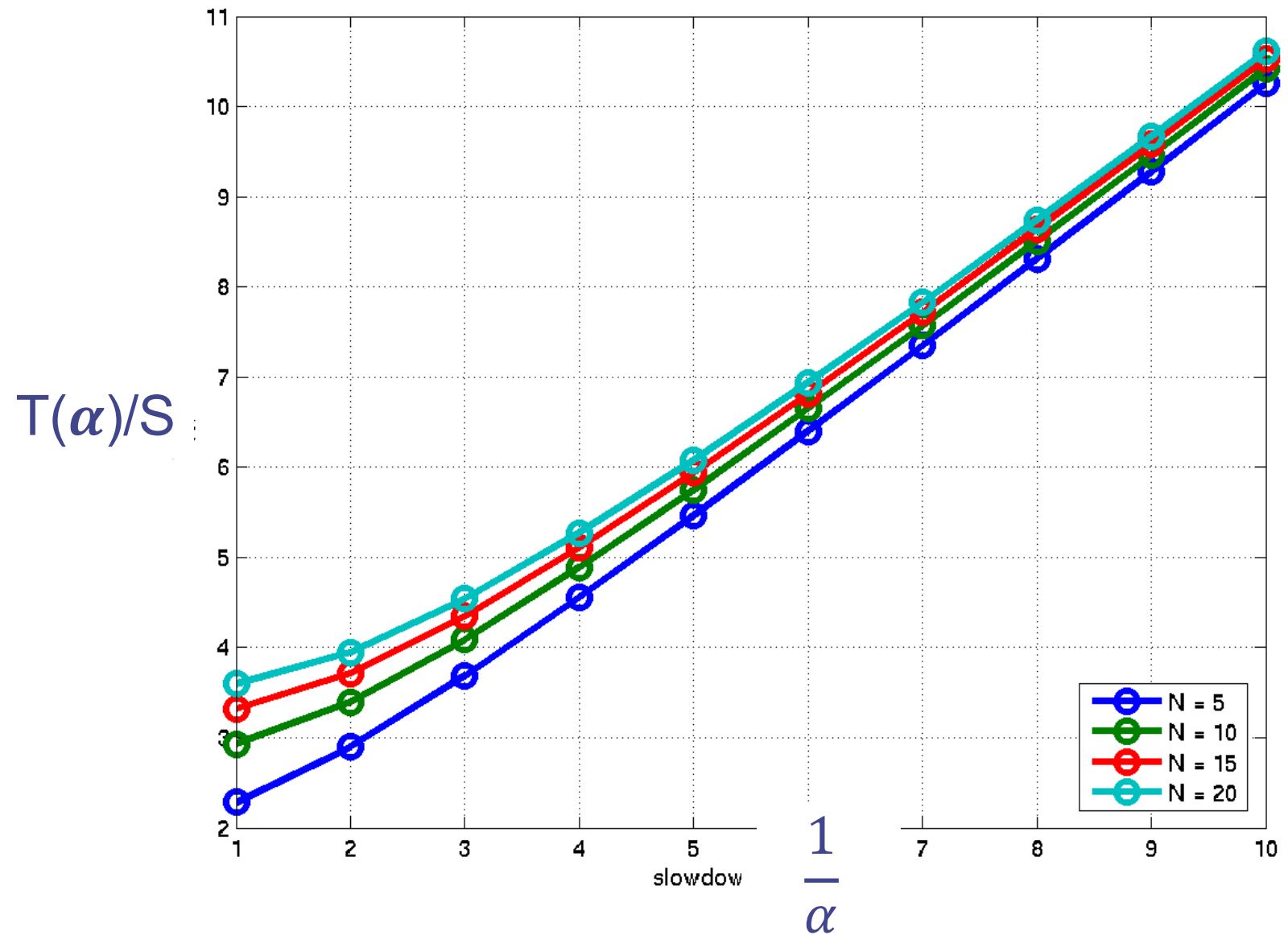
$$P(N) = \left[1 + \sum_{i=1}^{N-2} F(i) + \frac{1}{\alpha} F(1) + V\right]^{-1}$$

$$V = \alpha \sum_{j=1}^{N-1} \frac{1}{j} \sum_{i=j}^{N-1} F(i),$$

Note: When $\alpha = 1$,
 $T(\alpha) = H_N S$

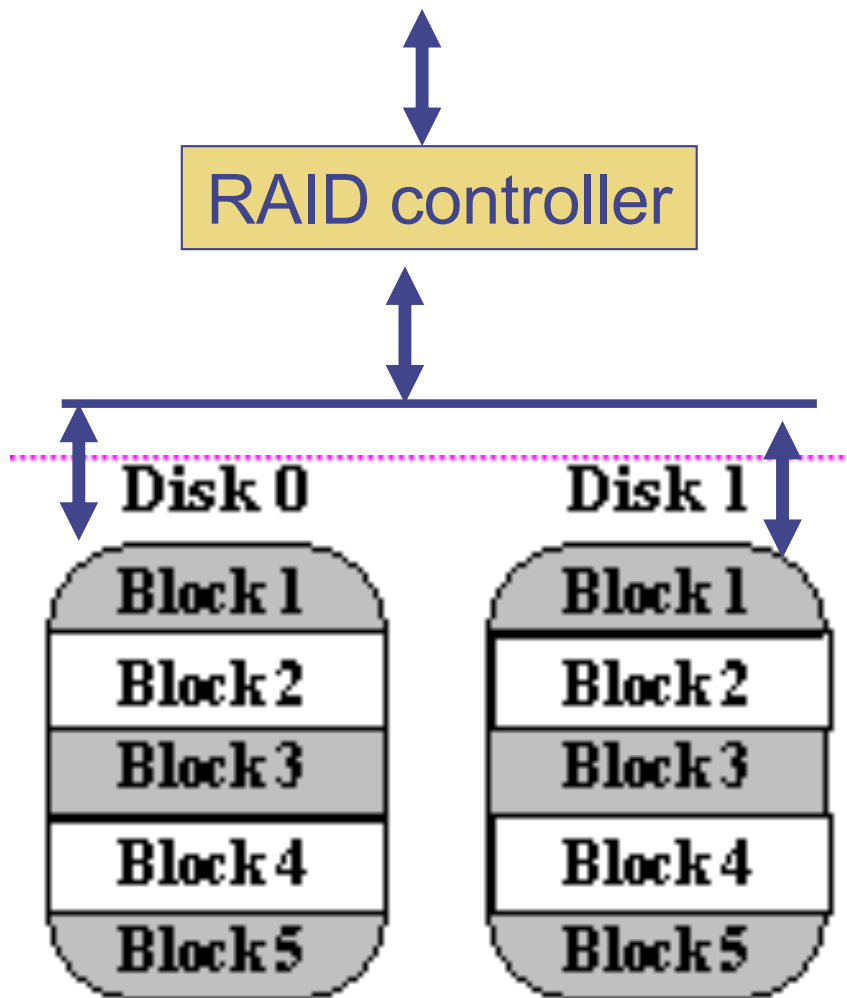
$$F(i) = \prod_{j=1}^{N-i-1} \frac{N-j}{N-j-1+\alpha}.$$

How $T(\alpha)/S$ varies with α ?



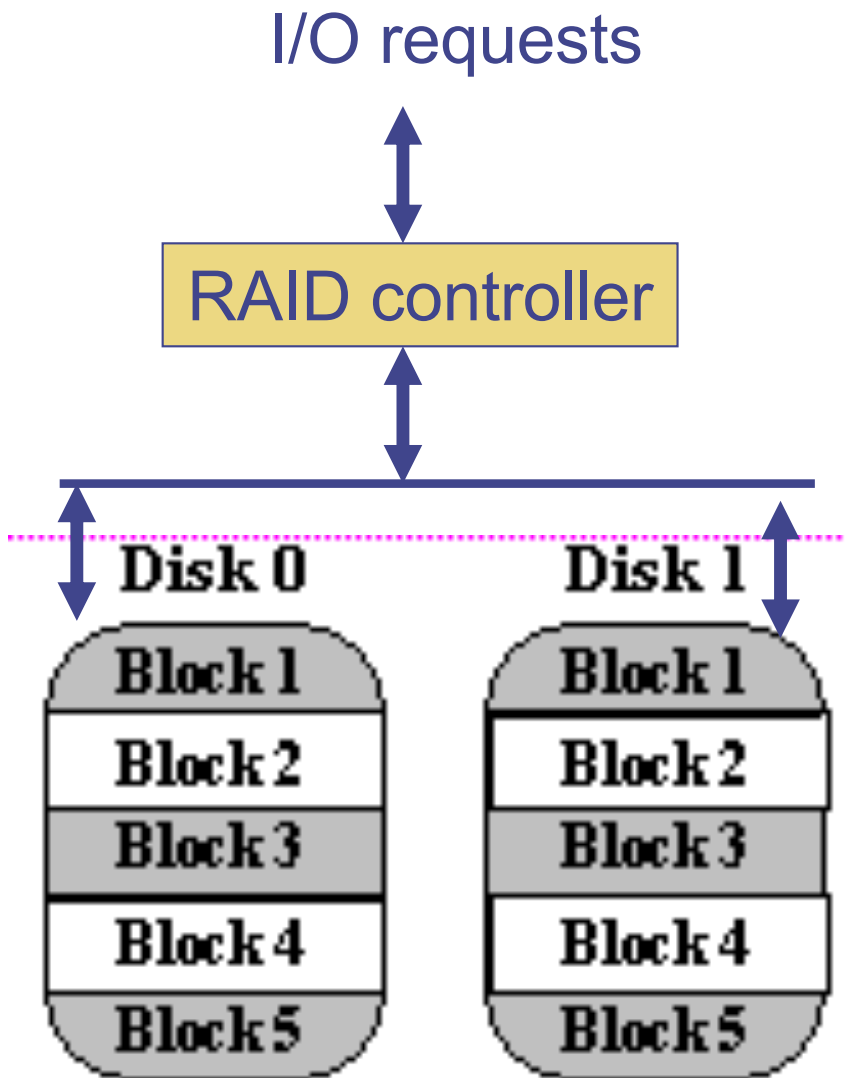
Disk array as an example of fork-join QNs

- Disk array, e.g. RAID (= Redundant Array of Independent Disks)



Example of RAID1
Mirrored disks

Fork-join in disk array



Example 1

Read a file in parallel

1st half of the file from Disk 0

2nd half of the file from Disk 1

Need to wait for both halves of the file before the next operation

Example 2

Write to disk.

Need to write to both disks (for consistency)

Need to wait for both disks to complete

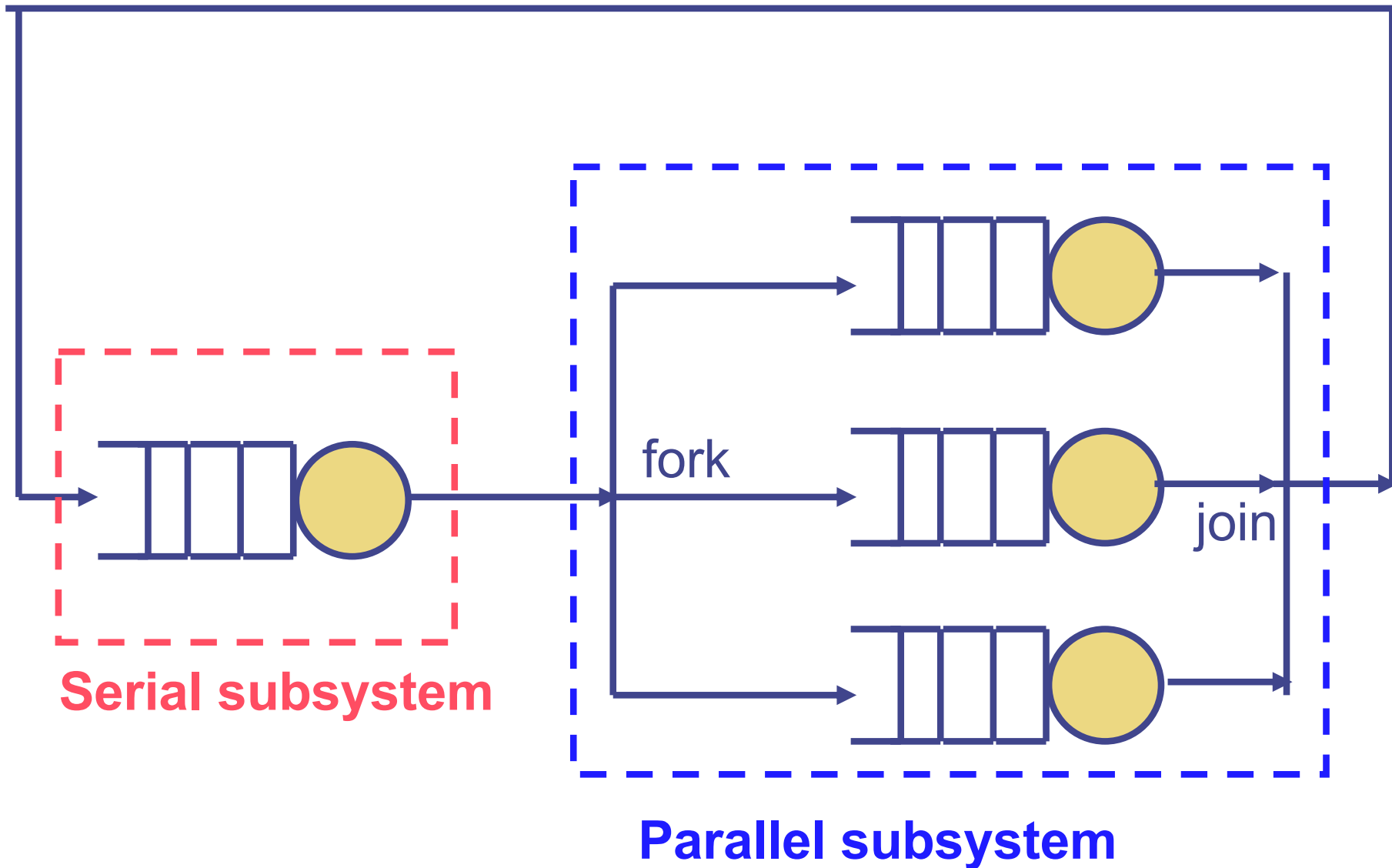
Parallel computing as an example of fork-join

- Computational clusters
 - Provide computing service to researchers, scientists etc.
 - Hundreds or even thousands of processors
 - Each processor consists of multiple cores
 - Each core is a computational unit
 - The CPU of your computer may have 2, 4, 6, 8 or 10 cores per processor
 - Processors in clusters may have 24, 48, .., cores per processor
 - You can use these cores in parallel
 - Example:
 - You want to perform 50 independent replications for your simulation
 - You can distribute these 50 simulations to different cores (fork)
 - Your job, which consists of 50 simulations, is complete when all 50 of them are finished (join)

Fork-join queueing networks

- Exact results are hard to come by
- Approximate solution methods are used

A Queueing network with a fork-join subsystem

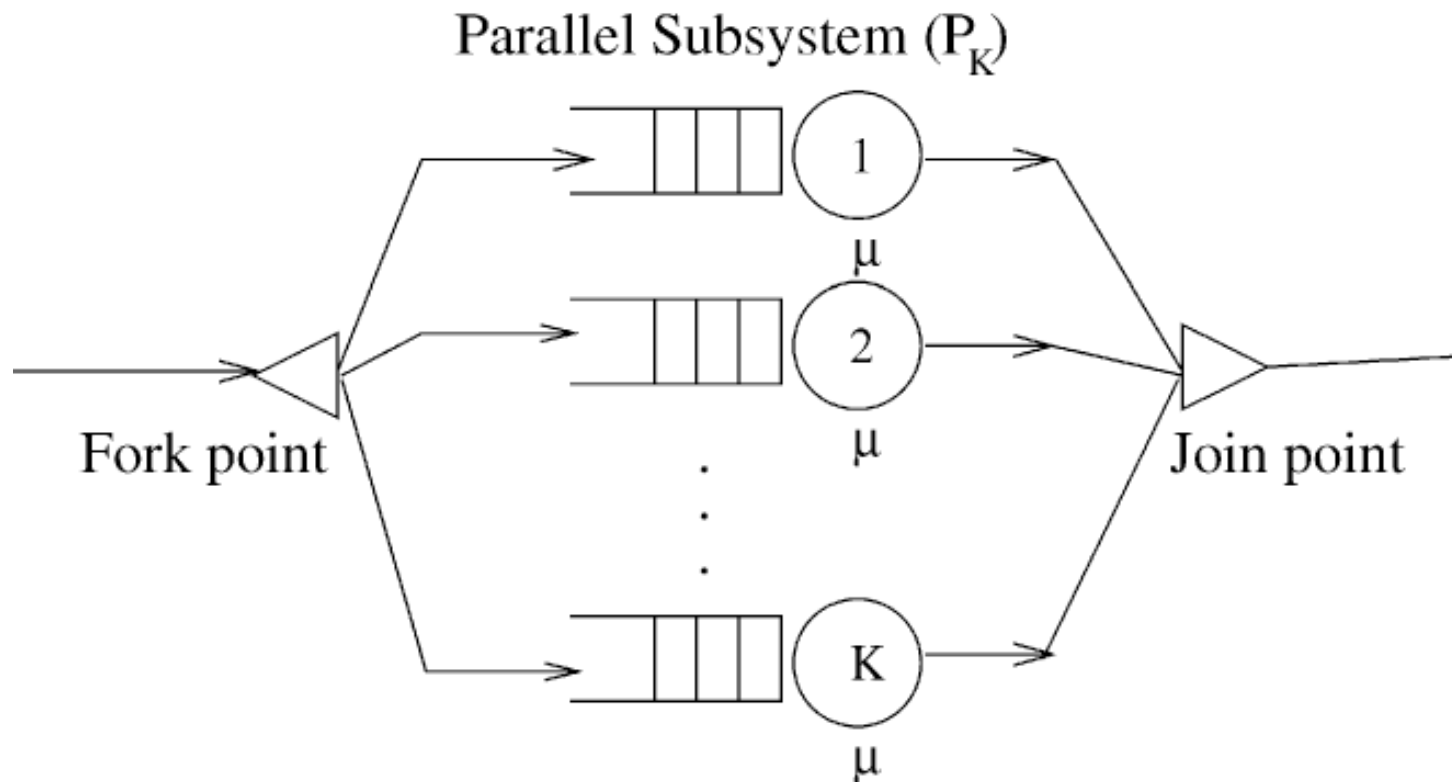


Approximate MVA for fork-join queueing networks

- For MVA with fork-join, the basic unit is a subsystem
 - A subsystem can be either a serial subsystem (= a device) or parallel one
 - A serial subsystem is a special case of parallel subsystem
 - In comparison, the basic unit for MVA before is a device

Arrival Theorem for Parallel Subsystems (1)

- Consider a parallel subsystem with k parallel service centres
- The average time each job requires at each service centre is S (exponentially distributed)



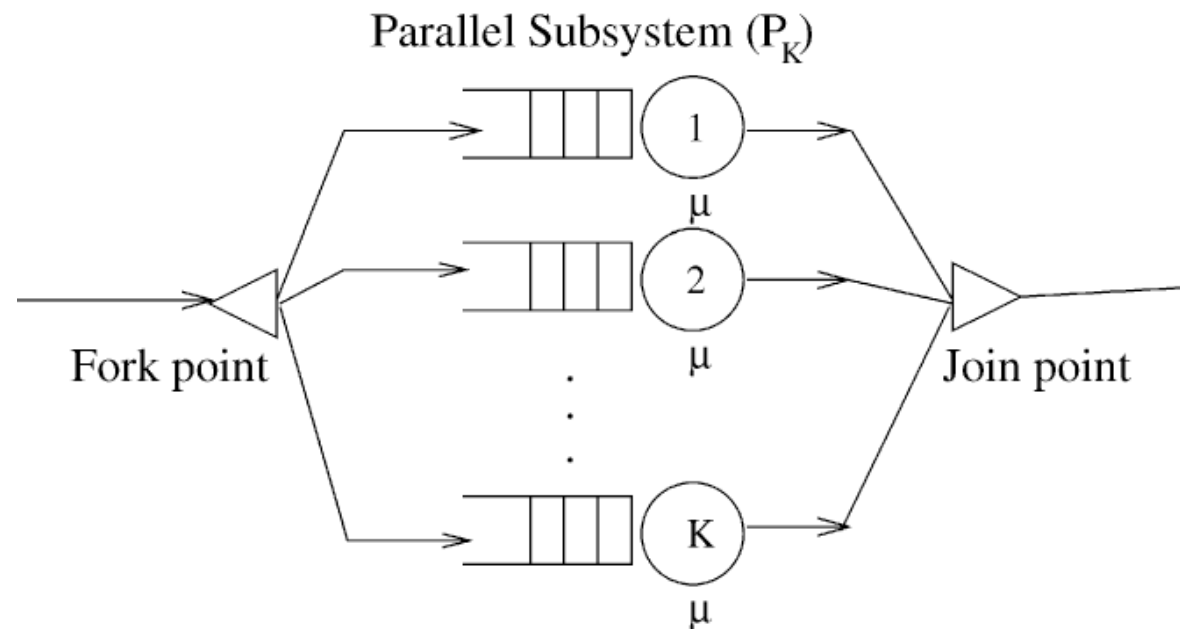
$$\mu = 1/S$$

Arrival Theorem for Parallel Subsystems (2)

When there are $n - 1$ jobs in the whole QN, the average number of jobs in the subsystem is z . When there're n jobs in the system



One of the
 n jobs (customers)



$$\text{Waiting time} = S \times z; \text{Service time} = S \times H_k$$
$$\Rightarrow \text{Response time} = S \times (H_k + z)$$

Note that if $k = 1$, the subsystem is serial and is identical to a device in MVA analysis that we have seen before.

$$\begin{aligned}\text{Response time} &= S \times (H_1 + z) \\ &= S \times (1 + z) \\ &\quad (\text{Since } H_1 = 1)\end{aligned}$$

This is the same arrival theorem that we've seen before.

Notation:

I = Number of subsystems in the QN

S_i = Avg. service time of a station in subsystem i

k_i = # parallel stations in subsystem i

$R_i(n)$ = Response time at subsystem i
when there're n jobs in the QN

$\bar{n}_i(n)$ = Avg. # of jobs at subsystem i
when there're n jobs in the QN

V_i = Visit ratio of subsystem i

MVA for fork-join systems:

Mean # jobs in each subsystem

$$\bar{n}_i(n-1)$$

(n-1) jobs in system

n jobs in system

$$R_i(n) = S_i \times (H_{k_i} + \bar{n}_i(n-1))$$

Mean response time of each subsystem

$$R_i(n)$$

$$X_0(n) = \frac{n}{R_0(n)} = \frac{n}{\sum_{i=1}^I V_i R_i(n)}$$

Throughput of the system

$$X_0(n)$$

$$\bar{n}_i(n) = V_i \times X_0(n) \times R_i(n)$$

Mean # jobs in each subsystem

$$\bar{n}_i(n)$$

Example

- A system consists of a processor and 2 disk arrays
- Disk arrays operate under synchronous workload
 - Transactions are blocked until I/O are completed

	Service demand	# parallel systems
Processor	0.01	1
Disk array 1	0.02	2
Disk array 2	0.03	3

What is the system response time when there are 50 transactions? How many transactions can the system have if the system response time should not exceed 1s?

Exercise

- The MVA algorithm on p.36 assumes that you have both visit ratios V_i and mean service time S_i available
- You may recall that service demand $D_i = V_i * S_i$
- Now, let us assume that you are only given the service demands D_i . That is, you know only D_i but you do not know V_i and S_i . How can you modify the MVA algorithm on p.35 so that it can work with knowing service demands only?

Summary

- Fork-join queues can be used to model web services, disk array, some parallel computing tasks
- Can analyse fork-join queues by using
 - Operational analysis
 - MVA provided that the service time at each parallel service centre is exponentially distributed

References (1)

- Web services
 - D. Mensace et al. Static and Dynamic Processor Scheduling Disciplines in Heterogeneous Parallel Architectures," *Journal of Parallel and Distributed Computing*, Vol. 28 (1), July 1995, pp. 1-18.
 - D. Mensace, "QoS Issues in Web Services," *IEEE Internet Computing*, November/December 2002, Vol. 6, No. 6.
 - D. Mensace, "Response Time Analysis of Composite Web Services," *IEEE Internet Computing*, January/February 2004, Vol. 8, No. 1
 - D. Mensace, "Composing Web Services: A QoS View," D. Menasce, *IEEE Internet Computing*, Vol. 8., No. 6, Nov/Dec 2004.
 - These papers can be downloaded from the course website (use your CSE password)
 - We didn't cover the last paper but it's well worth a read.
- Derivation of Markov chain on pp. 22-24 is further explained in the file *forkjoin_mc.pdf*

References (2)

- Fork-join MVA
 - Menasce et al., "Performance by desing". Section 15.6.
- Addition references outside the scope of this course
 - Tutorial on RAID <http://www.slcentral.com/articles/01/1/raid/>